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CHE 230: Molecular Systems in Chemical Engineering Homework 3

Due: 4/01/2026

CHE 230 Time:11:59pm

1. a. Derive the lever rule. (**5 points**)

It depends on what Phases we have and what is being asked

$$f_\alpha = f_V X + f_L (1-X)$$

$$f_\alpha = f_V X + f_L - f_L X$$

$$f_\alpha - f_L = f_V X - f_L X$$

$$f_\alpha - f_L = X (f_V - f_L)$$

$$\boxed{\frac{f_\alpha - f_L}{f_V - f_L} = X}$$

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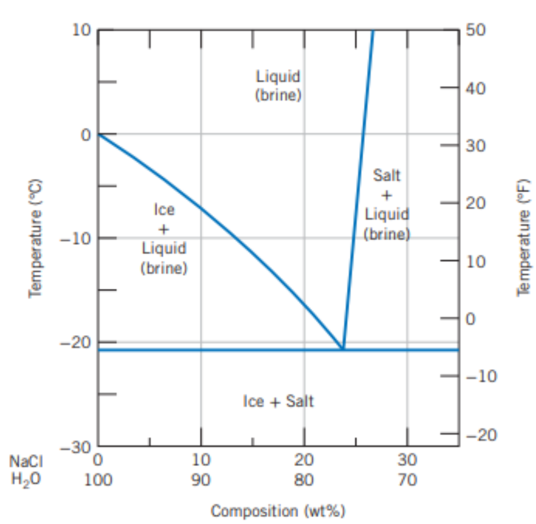
There are other forms of this such as:

$$X = \frac{C_0 - C_L}{C_\beta - C_L} \quad \text{or} \quad X = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} \quad \text{or} \quad X = \frac{C_0 - C_L}{C_\alpha - C_L}$$

b. The following is a portion of the H_2O–NaCl phase diagram: (**5 points)**



(i) Using this diagram, briefly explain how spreading salt on ice that is at a temperature below 0°C (32°F) can cause the ice to melt.

(ii) At what temperature is salt no longer useful in causing ice to melt?

i) The freezing Point of Pure water is 0°C whe NaCl is added it lowers the freezing Point of the water Causing brine to form. meaning at lower temperatures adding NaCl to the ice Puts the system in the ice+liquid (brine) region where both Phases liquid brine and Solid ice and the liquid brine needs to exist in order for the ice to melt.

ii) at $T=-21°C$ or $-6F$ Salt is no longer useful in Causing the ice to melt because below that level the liquid brine no longer exists.

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2. 90 wt% Ag-10 wt% Cu alloy is heated to a temperature within the β + liquid phase region. If the composition of the liquid phase is 85 wt% Ag, determine: (**14 points**)

(a) The temperature of the alloy → T ≅ 850°C

(b) The composition of the β phase → β = 95 wt%

(c) The mass fractions of both phases

$$X = \frac{C_0 - C_L}{C_\beta - C_L}, \quad C_0 = 90 \text{ wt\%}, \; C_L = 85 \text{ wt\%}, \; C_\beta = 95 \text{ wt\%}$$

$$X = \frac{90-85}{95-85} = \frac{5}{10} \Rightarrow X = 0.50$$

$X_L = 0.50$

$X_\beta = 0.50$

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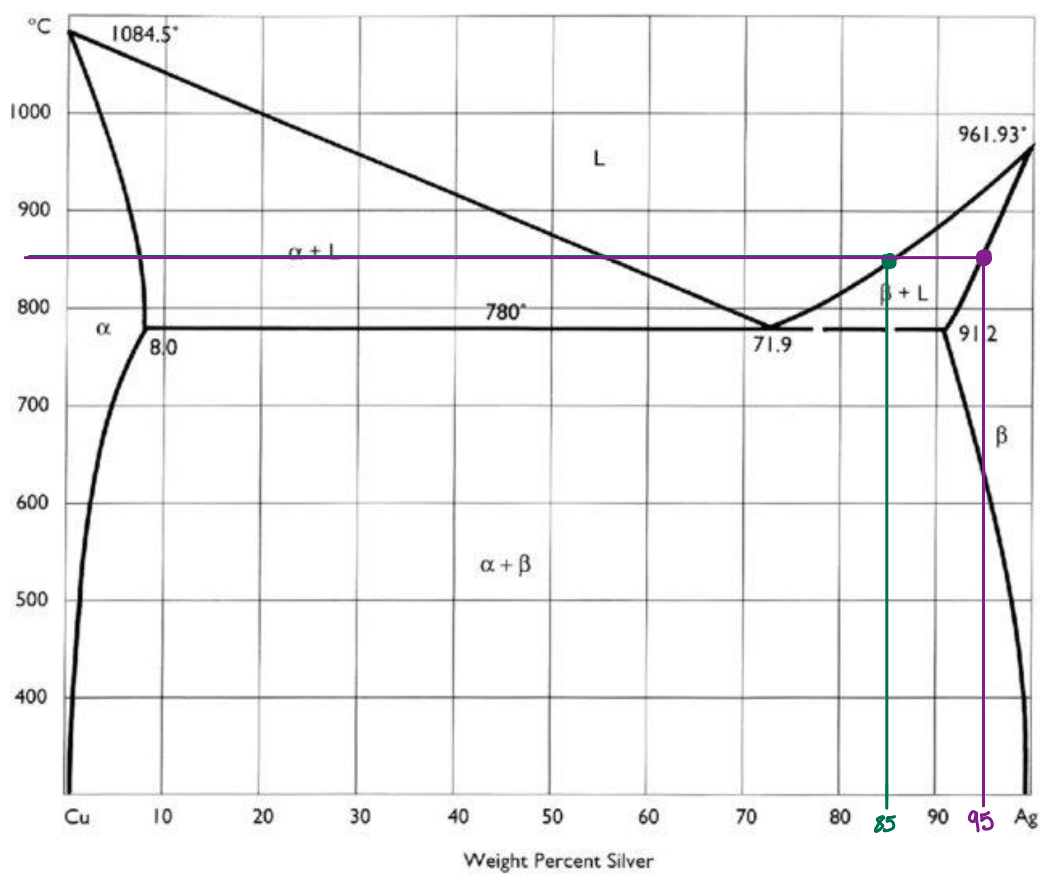

3. a. Explain the phenomenon of temperature change from a molecular perspective and why it is that some materials have higher heat capacity than others (**6 points)**.

Materials are made of small atoms or particles so when heat is added to a material it causes it's particles to gain energy and starts to move around in faster motion, when particles starts moving and gaining energy it causes the temperature to change or to rise and when heat is removed this causes the particles to stop moving as fast and there will be no energy that is being gained which cause the temperature of the material to drop.

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Materials have higher heat capacity than others because heat capacity tells us how much heat is needed to raise the temperature of the material to a certain level meaning materials with higher heat capacity requires more heat and can absorb more energy in order to rise it's temperature. This occurs because materials can have stronger or more complicated bonds or because materials can store energy differently or because particles can use this energy of motion in different ways, vibration, rotation, etc.

b. Explain why a brass lid ring on a glass canning jar loosens when heated (**4 points**).

When the brass lid ring gets heated it thermally expand meaning it changes size when temperature changes so therefore the lid will expand causing it to fit less tightly around the glass so it loosens up.

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c. A brass rod is to be used in an application requiring its ends to be held rigid. If the rod is stress-free at room temperature (20°C or 68°F), what is the maximum temperature to which the rod may be heated without exceeding a compressive stress of 172 MPa (25,000 psi)? $\rightarrow 172\ MPa \times \frac{10^6\ Pa}{1\ MPa} = 172 \times 10^6\ Pa$

Assume the following properties for brass:

- Modulus of elasticity (E) = 100 GPa (14.6 x 10^6 psi) $\rightarrow 100\ GPa \times \frac{10^9\ Pa}{1\ GPa} = 100 \times 10^9\ Pa$
- Linear coefficient of thermal expansion (α) = 20.0 x 10^{-6} °C^{-1} (11.1 x 10^{-6} °F^{-1}) (**8 points)**

$$\sigma_T = E \alpha_\ell \Delta T$$

$$(172 \times 10^6\ Pa) = (100 \times 10^9\ Pa)(20 \times 10^{-6} \tfrac{1}{C})(T_{max} - 20°C)$$

$$86°C = T_{max} - 20°C \Rightarrow T_{max} = 106°C$$

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* α for #4 I had to google
wasn't sure where to get it
from since it wasn't given *

4. An aluminum wire 10 m (32.8 ft) long is cooled from 38°C to -1°C (100°F to 30°F). How much change in length will it experience? (**6 points**)

$\Delta L = \alpha L_0 \Delta T$

$\Delta L = (23\times10^{-6} \frac{1}{C})(10m)(-1-38)°C$

$\Delta L = -0.00897m$

or

It decreased by 8.97×10^{-3} m

✓ 6

5. Compute the electrical conductivity of a cylindrical silicon specimen 5.1 mm (0.2 in.) in diameter and 51 mm (2 in.) in length in which a current of 0.1 A passes in an axial direction. A voltage of 12.5 V is measured across two probes that are separated by 38 mm (1.5 in.).

(b) Compute the resistance over the entire 51 mm (2 in.) of the specimen (**8 points**).

a) $\sigma = \frac{J}{\mathcal{E}}$, $\mathcal{E} = \frac{V}{l}$, $J = \frac{I}{A}$ $\Rightarrow \sigma = \frac{Il}{VA}$

$A = \pi r^2 \Rightarrow \pi\left(\frac{5.1}{2}mm\right)^2 \Rightarrow A = 20.43\ mm^2$

$\sigma = \frac{(0.1A)(38mm)}{(12.5V)(20.43mm^2)} \Rightarrow \sigma = 0.01488\ S/mm \Rightarrow \sigma = 14.88\ S/m$

✓ 4

b) $R = \frac{V}{I} \Rightarrow R_{38} = \frac{12.5}{0.1} \Rightarrow R_{38} = 125$ $\quad R_{51} = R_{38}\left(\frac{51}{38}\right) \Rightarrow R_{51} = 125\left(\frac{51}{38}\right) \Rightarrow R_{51} = 167.76\Omega$

✓ 4

6. An n-type semiconductor is known to have an electron concentration of 3×10^{18} m^{-3}. If the electron drift velocity is 100 m/s in an electric field of 500 V/m, calculate the conductivity of this material **(4 points)**.

$\mu_e = \frac{\sigma}{n|e|}$, $\mu_e = \frac{V_d}{E}$

$\mu_e = \frac{100\,m/s}{500\,V/m} \Rightarrow \mu_e = 0.2\,\frac{m^2}{Vs}$

$\sigma = (3 \times 10^{18}\,m^{-3})(1.6 \times 10^{-19}\,C)(0.2\,\frac{m^2}{Vs}) \Rightarrow \sigma = 0.096\,S/m$

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